

Rensselaer Polytechnic Institute
Department of Electrical, Computer, and Systems Engineering
ECSE-4040 – Digital Electronics
Spring Term 2026

Lab. 6- Delays and Sequential Circuits

Issued: 2/25/2026

Due: 2/27/2026

Given

Below is a table of process parameters for a generic 0.13 μm technology.

0.13 μm Process technology parameters

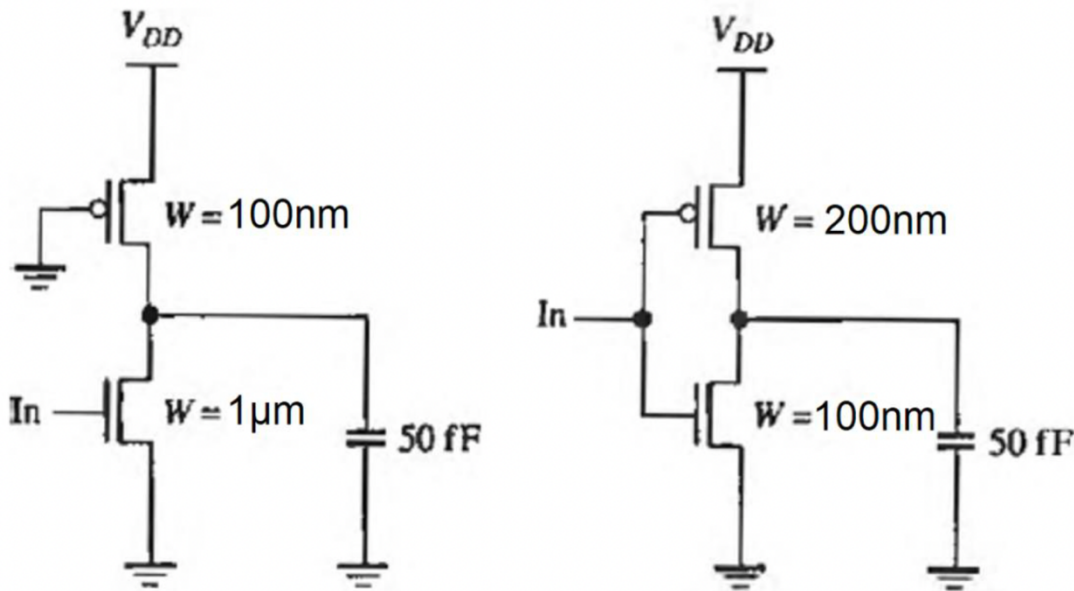
Typical Technology Parameters	Symbol	NMOS	PMOS
Gate thickness	t_{ox} [Å]	22	22
Poly doping level	N_D [cm^{-3}]	3×10^{20}	—
Poly doping level	N_A [cm^{-3}]	—	3×10^{20}
Substrate doping level	N_A [cm^{-3}]	3×10^{17}	—
N-well doping level	N_D [cm^{-3}]	—	3×10^{17}
Number of surface state charges	N_{ss} [cm^{-2}]	6×10^{11}	6×10^{11}
Electron Mobility	μ_n [cm^2/Vs]	270	—
Hole Mobility	μ_p [cm^2/Vs]	—	70
Critical Field x Length	$E_C L$ [V]	0.6	2.4
Saturation velocity	v_{sat} [cm/s]	8×10^6	8×10^6
Gamma	γ [$\text{V}^{1/2}$]	0.2	0.2
Oxide capacitance	C_{ox} [F/cm^2]	1.6×10^{-6}	1.6×10^{-6}
Threshold voltage	V_T [V]	0.4	− 0.4
2 x Fermi voltage	$2 \phi_F $ [V]	0.88	0.88
Channel length modulation	λ [V^{-1}]	0.2	0.2

Question 1: Based on Problem 5.12 (D.A. Hodges, H.G. Jackson, and R.A. Saleh, *Analysis and Design of Digital Integrated Circuits*, 3rd Ed., McGraw-Hill, 2004.)

Objectives:

- Power regions and calculation

Given the circuits below. $V_{DD} = 1.2V$ (always assume this even if not given).



- a. Sketch the VTC of both circuits. No calculations are necessary. Just plot the rough differences.
 - i. Label on the plot the following: i. Regions of static current.
 - ii. Regions of short-circuit current.
 - iii. Regions of subthreshold current.
- b. Calculate the dynamic power dissipation of both circuits at 2 GHz if $\alpha=1$. Assume transistor capacitance is negligible.

Question 2: Based on Problem 5.16 (D.A. Hodges, H.G. Jackson, and R.A. Saleh, *Analysis and Design of Digital Integrated Circuits*, 3rd Ed., McGraw-Hill, 2004.)

Objectives:

- SR Latch Design

Given: $V_{DD} = 1.2V$. $L = 100nm$.

Design an SR latch in $0.13\mu m$ CMOS using NAND gates to deliver a delay of 400 ps from S to Q and from R to $\bar{Q}$. Assume that the total load to be driven by Q and $\bar{Q}$ is 100fF.

HINT: Calculate the R_{eff} required to meet τ_{PHL} and τ_{PLH} requirements for both NAND gates. From this determine the required W_P and W_N sizing.

Report Submission:

Submit your work on LMS under **Week 7/Lab 6**.

- Show your work.
- Include images of your hand calculations with work and final results.
- Include screenshots of LTspice with all circuit schematics with all output plots, **clearly showing all hand calculated parameters**.
- Due date: Friday at 11:59pm.

